

01 CONTEXT AND HYPOTHESIS

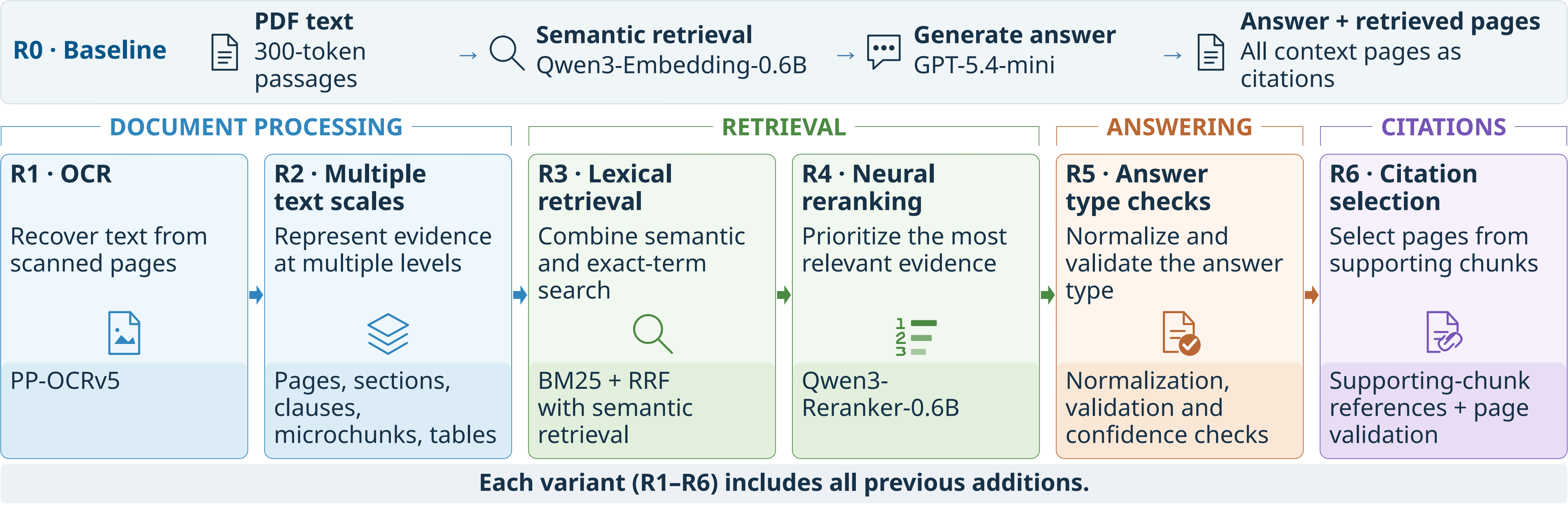
Retrieval-augmented generation (RAG) grounds language-model answers in external documents [1]. Legal QA needs correct answers and precise, complete source-page citations [2, 3]. PDFs may lack searchable text; evidence spans different text scales and may require exact legal terms. Retrieving more passages can also introduce irrelevant pages.

How can a legal RAG pipeline improve answer quality while keeping source-page citations both precise and complete?

H1 · Retrieval Multiple text scales, hybrid search and reranking will improve answers and evidence retrieval over R0.	H2 · Scan robustness Recovering text from scanned documents will reduce the performance loss caused by the missing text layer.	H3 · Citation selection Selecting pages after answer generation will improve citation precision while retaining most of the relevant evidence.
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02 METHODOLOGY

Data: 30 DIFC legal PDFs (590 pages), 100 questions: 70 structured-answer and 30 free-text. Reference source pages are available for 95 questions. In the mixed corpus, 10 PDFs (107 pages) become synthetic scans: page images at 300 DPI with the text layer removed.

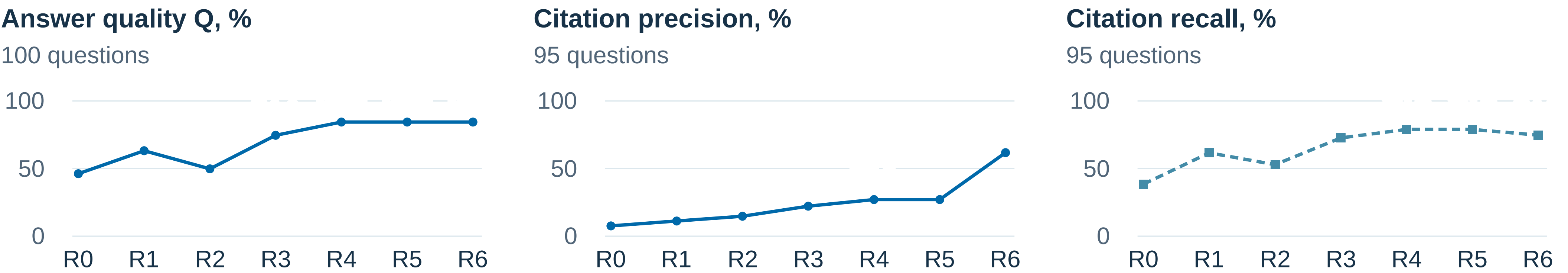


Answer generation: A fixed GPT-5.4-mini configuration is used across variants, so changes in performance reflect modifications to the surrounding pipeline rather than a change of the generating LLM.

Answer quality <i>Is the generated answer correct?</i> $Q = 0.7 \times \text{structured score} + 0.3 \times \text{free-text score}$	Citation precision <i>Are the cited pages actually relevant?</i> Share of cited pages that support the answer.	Citation recall <i>Did the system cite all required evidence?</i> Share of annotated supporting pages that were cited.
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03 RESULTS

R0 to R6 on the mixed corpus



OCR recovers performance on scanned documents

Original corpus · R0	65.4
Mixed corpus (original + synthetic scans) · R0	46.2
Mixed corpus + OCR · R1	63.2

0 50 100%

Converting 10 documents into synthetic scans reduces answer quality from 65.4% to 46.2%. OCR restores most of this loss, reaching 63.2%.

Hypothesis evaluation

H1 · Partially supported. Multiple scales alone reduce Q; lexical retrieval and reranking improve Q and recall.

H2 · Supported. OCR restores most of the loss caused by removing the text layer.

H3 · Supported with a trade-off. Precision rises from 27.1% to 61.8%; recall falls from 78.9% to 74.7% (−4.2 pp).

Limitations

one DIFC corpus and 100 questions; one run per condition with no significance testing; LLM-based evaluation of free-text answers.